

Chapter 59: Hematological Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2724–2778 of the PDF. High-yield numbers are in **bold**. The seven tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 3 figures are included from the PDF.

Big picture: Pregnancy's physiological hemodilution, falling platelets and rising clotting factors can both hide and mimic disease. **Iron deficiency and acute blood loss** cause most anemia; **gestational thrombocytopenia** causes most low platelets. The dangerous conditions to recognize are **sickle-cell disease**, **thrombotic microangiopathies (vs HELLP)**, and inherited bleeding disorders (**vWD, hemophilia carriers**).

1. Anemias — Definition and Evaluation

- **CDC definition** (iron-supplemented women, 5th percentile): **Hb <11 g/dL in the 1st and 3rd trimesters; <10.5 g/dL in the 2nd trimester**. (Not derived from a US population.)
- **Parkland data (480 iron-sufficient women): Hct <30% is a reasonable definition of anemia.**
- Mechanism of the physiological fall: plasma volume expands more than red cell mass — **greatest disproportion in the 2nd trimester**; late in pregnancy plasma expansion stops while Hb mass keeps rising.
- US prevalence **3–38%**.
- **Initial workup (moderate anemia):** Hb, Hct, red cell indices, serum iron or ferritin, **peripheral smear**, and a **sickle-cell prep** if of African lineage.

Table 59-1. Hematocrit Values in Pregnancy (percent)

	5th percentile	50th percentile	75th percentile
1st trimester	33.0	37.5	41.2
2nd trimester	30.5	35.7	39.2
Predelivery	30.7	36.5	40.5

Data from 480 iron-sufficient women at Parkland Hospital (Zofkie, 2020).

Table 59-2. Causes of Anemia During Pregnancy

Acquired	Hereditary
Iron-deficiency anemia	Thalassemias
Acute blood-loss anemia	Sickle-cell hemoglobinopathies
Anemia of chronic disease	Other hemoglobinopathies
Megaloblastic anemia	Hemolytic anemias
Hemolytic anemias	
Aplastic or hypoplastic anemia	

Effects on pregnancy outcome

- **Mild anemia (Hb 9.0–10.9 g/dL)** in 12% of >500,000 Canadian women → **2.5× transfusion risk**. **Moderate anemia** → ↑ FGR, low 5-min Apgar, perinatal mortality.
- Correcting iron deficiency ↓ transfusion at delivery.

- **Paradox: high Hb is also bad** — reflects poor plasma volume expansion. Hb **>3 SD above mean at 12 or 18 wk** → **1.3–1.8× FGR**. Placental weight correlates negatively with Hb.
- ⚠ This does **not** justify withholding iron.

2. Iron-Deficiency Anemia

- **Two most common causes of anemia in pregnancy/puerperium: iron deficiency and acute blood loss.**
- Singleton pregnancy needs **~1000 mg iron** (more with multiples) — exceeds most women's stores.
- **Fetus gets its iron regardless** → newborn of a severely anemic mother is **not** iron-deficient anemic. Neonatal stores depend on maternal status and **timing of cord clamping**.
- **Diagnosis:**
 - Hypochromia + microcytosis (less prominent in pregnancy); **MCV <80 fL**
 - **Serum ferritin <10–15 confirms** (ferritin normally falls in pregnancy)
 - Hepcidin falls in pregnancy (hepcidin blocks ferroportin iron export)
- **Prophylaxis (WHO): 30–60 mg elemental iron + 400 µg folic acid daily.**
- **Treatment: ~200 mg elemental iron daily** (ferrous sulfate, fumarate or gluconate).
 - Can't take oral iron → **IV iron; iron sucrose is safer than iron dextran.**
 - Response = **↑ reticulocytes**; Hb rises more slowly than in nonpregnant women (larger plasma volume).
 - Oral and parenteral iron give **equivalent** Hb and ferritin rises.

The extracted text gives ferritin as "10 to 15 mg/L" (should be µg/L, i.e., ng/mL) and says "ferrous sucrose" (should be iron sucrose).

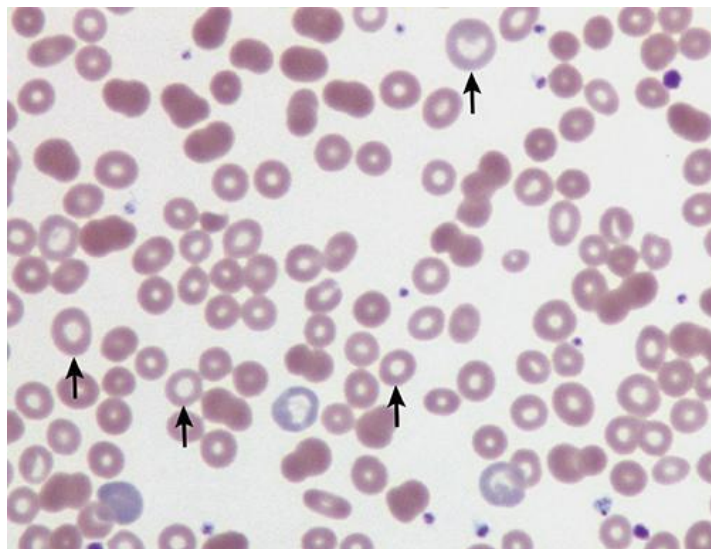


Figure 59-1. Iron-deficiency anemia smear: scattered microcytic, hypochromic red cells with central pallor (arrows).

Anemia from acute blood loss

- Early pregnancy: **abortion, ectopic, molar pregnancy**. Postpartum: obstetric hemorrhage (massive hemorrhage → Ch 44).
- **Hb ~7 g/dL, hemodynamically stable, ambulating without symptoms, not septic** → **NO transfusion**; give oral iron for **≥3 months**.

Anemia of chronic disease

- Inflammatory cytokines restrict erythropoiesis and shorten RBC life; **hepcidin ↑** blocks enterocyte iron export.
- **Slightly hypochromic/microcytic, low transferrin saturation, HIGH ferritin.**

- With iron deficiency, the most common anemia worldwide. May appear or worsen in pregnancy as plasma expands.
- Causes: **chronic renal insufficiency (most common at Parkland)**, IBD, connective tissue disease, granulomatous infection, malignancy.
- Renal disease: red cell mass expansion is inversely related to renal impairment while plasma expansion is normal → worse anemia; some EPO deficiency.
- **Treatment:** ensure iron stores; **recombinant erythropoietin** ($\pm$ IV iron), considered when **Hct ~20%** in renal insufficiency. Side effects: **hypertension**, red cell aplasia, anti-EPO antibodies.

	Iron deficiency	Anemia of chronic disease
MCV / color	Microcytic, hypochromic	Slightly microcytic/hypochromic
Ferritin	Low (<10–15)	High
Transferrin saturation	Low	Low
Hepcidin	↓	↑

3. Megaloblastic Anemia

Folic acid deficiency

- Impaired DNA synthesis → large cells with **nuclear maturation arrested** while cytoplasm matures.
- **In pregnancy almost always folate deficiency** (formerly "pernicious anemia of pregnancy"): diets lacking **green leafy vegetables, legumes, animal protein**; anorexia worsens it.
- Other causes: malabsorption (tropical sprue, jejunal resection, gastrectomy, Crohn), hemolytic anemias, malignancy, antifolate drugs.
- **Requirement: 50–100 μ g/d nonpregnant → 400 μ g/d in pregnancy.**
- **Sequence of findings:** low plasma folate (earliest) → **hypersegmented neutrophils + macrocytes** → nucleated RBCs, megaloblastic marrow → severe anemia $\pm$ thrombocytopenia/leukopenia.
- **Fetus is not anemic** despite severe maternal anemia.
- **Treatment: oral folic acid 5–15 mg/d + iron** + nutritious diet → **reticulocytosis and correction of WBC/platelets by 4–7 days.**
- Folate also prevents neural-tube defects (Ch 15).

Vitamin B12 deficiency

- B12 levels fall in pregnancy ($\downarrow$ **transcobalamins**), but B12-deficient megaloblastic anemia is **rare**.
- Most likely after **gastric resection: total gastrectomy → cyanocobalamin 1000 μ g IM monthly**; partial gastrectomy — check levels.
- Other causes: Crohn, ileal resection, drugs, bacterial overgrowth.
- **Addisonian pernicious anemia** (no intrinsic factor, autoimmune) usually starts **after 40** → uncommon in pregnancy.

4. Hemolytic Anemias

Autoimmune hemolysis

- **Direct and indirect Coombs positive. Warm antibodies 80–90%**, cold, or mixed.
- Primary (idiopathic) or secondary: lymphoma/leukemia, connective tissue disease, infection, chronic inflammation, drugs. **Cold agglutinins:** *Mycoplasma pneumoniae*, EBV.
- IgM or IgG anti-RBC antibodies. **With thrombocytopenia = Evans syndrome.**

- Hemolysis can accelerate markedly in pregnancy.
- **First line: low-dose rituximab 100 mg weekly × 4 + prednisone.** Transfusion is difficult; **warm donor cells** to body temperature (cold agglutinins). Fetus rarely involved.

Drug-induced hemolysis

- Usually mild; resolves on withdrawal.
- **High-affinity hapten** (drug on RBC protein; IgM antibodies): **penicillin, cephalosporins** — severe Coombs+ hemolysis from **cefotetan** prophylaxis in 7 women; **α-methyldopa** similar.
- **Low-affinity haptens:** probenecid, quinidine, rifampin.
- Common mechanism: **G6PD deficiency**.

Pregnancy-associated hemolysis

- Unexplained severe hemolysis in **early pregnancy**, resolves months postpartum; no immune mechanism or RBC defect found; fetus may have transient hemolysis. **Corticosteroids often effective.** Can recur each pregnancy.
- Pregnancy-specific: **HELLP** (mild microangiopathic hemolysis), **AFLP** (moderate–severe hemolysis).

Paroxysmal nocturnal hemoglobinuria (PNH)

- Acquired **hemopoietic stem cell clonal** disorder; mutated X-linked **PIG-A** gene → defective GPI anchor proteins → RBCs/granulocytes **susceptible to complement lysis**.
- Hemoglobinuria irregular, **not necessarily nocturnal**; triggered by transfusion, infection, surgery.
- **VTE in ~40%**; renal failure, hypertension, **Budd-Chiari**. → **Prophylactic anticoagulation**.
- **Treatment: eculizumab** (complement inhibitor; apparently safe in pregnancy). Definitive: **bone marrow transplant**. Median survival **10 years**.
- **Pregnancy:** complications in up to $\frac{3}{4}$; past maternal mortality **10–20%**; mostly **postpartum**; **half develop VTE**. On eculizumab (75 pregnancies): no maternal deaths, **stillbirth 4%**.

Bacterial toxins

- **Most fulminant acquired hemolysis in pregnancy: *Clostridium perfringens* exotoxin or group A β-hemolytic streptococcus.**
- Gram-negative endotoxin (e.g., **pyelonephritis**) → mild–moderate hemolysis; recovers after infection.

Inherited red cell membrane defects

- **Hereditary spherocytosis**, pyropoikilocytosis, ovalocytosis — among the most common hemolytic anemias in pregnancy.
- Usually **autosomal dominant, variably penetrant α- and β-spectrin deficiency**; also ankyrin, band 3, 4.1, 4.2 defects.
- **Diagnosis: abnormal cells on smear + ↑ osmotic fragility.**
- Crises: accelerated hemolysis (enlarged spleen); **parvovirus B19** suppresses erythropoiesis (aplastic crisis). Splenectomy for severe disease.
- **Pregnancy:** generally good. **Folic acid 4 mg/d.** Parkland: Hct 23–41% (mean 31), reticulocytes 1–23%; 8 of 23 women miscarried; 4 of 42 preterm, none FGR; infection worsened hemolysis (3 transfused).
- Newborn may have **hyperbilirubinemia and anemia**; fetal anemia uncommon.

Red cell enzyme deficiencies

- Hereditary **nonspherocytic** anemia; mostly **autosomal recessive**; crises triggered by drugs or infection.
- **Pyruvate kinase deficiency: 1 in 10,000**; transfusion → iron overload (watch myocardium); homozygous fetus → **hydrops**.

- **G6PD deficiency:** >200 variants; **X-linked**.
 - **~2% of African-American women homozygous (A variant); 10–15% heterozygous.** Lyonization → variable enzyme activity.
 - Hemolysis usually episodic; young RBCs have more enzyme, so anemia self-corrects once the trigger is removed.
 - Triggers: **pyelonephritis, pneumonia**, drugs — e.g., **nitrofurantoin (Macrochantin)** used for pyelonephritis suppression.
 - Offer preconception and genetic counseling ± partner testing.

5. Aplastic and Hypoplastic Anemia

- **Pancytopenia + markedly hypocellular marrow** (↓ committed stem cells). Cause found in **~1/3**: immune, drugs, chemicals, infection, irradiation, leukemia, inherited (**Fanconi, Diamond-Blackfan**).
- Treatment: immunosuppression; **eltrombopag** in some nonresponders. **Definitive = bone marrow transplant** (~¾ good response/long-term survival); cord blood stem cells are a source.
- **Pregnancy:** rare; **half diagnosed during pregnancy**; complication rate **79%** (19 pregnancies, no deaths). Pregnancy-induced hypoplasia improves after delivery/termination and may recur.
- **Diamond-Blackfan** (pure red cell hypoplasia): **40% familial, autosomal dominant**; responds to glucocorticoids; ⅔ of pregnancies had placental-vascular problems (miscarriage, preeclampsia, preterm, FGR, stillbirth).
- **Gaucher disease:** autosomal recessive **acid β -glucosidase** deficiency; anemia + thrombocytopenia worsen in pregnancy; **enzyme replacement (alglucerase, imiglucerase)**.
- **Main risks: hemorrhage and infection.** ↑ Preterm labor, preeclampsia, FGR, stillbirth.
- **Management:** infection surveillance + prompt antibiotics; granulocyte transfusion only during infection; **transfuse RBCs to keep Hct >20%**; platelets for hemorrhage. Mortality historically **~50%**, now lower.

After bone marrow transplantation

- Fertility ↓, but outcomes generally good; ~half of live births complicated by preterm delivery or hypertension; 20% had ↓ hemopoiesis needing transfusion.
- Normal pregnancy erythropoiesis and volume expansion. **Cell-free DNA results must account for donor DNA.**

6. Polycythemias

- **Secondary** (usual): chronic hypoxia — **cyanotic heart disease, chronic lung disease**; heavy smokers with bronchitis (Hct **55–60%**). Severe polycythemia → poor pregnancy outcome.
- **Polycythemia vera:** clonal myeloproliferative disorder; **virtually all JAK2 mutation**; hyperviscosity, **thrombosis**.
 - Nonpregnant: hydroxyurea or ruxolitinib.
 - **Pregnancy: high fetal loss → aspirin improves outcome; LMWH if prior VTE; interferon- α if cytoreduction needed.**

7. Hemoglobinopathies

Sickle-cell pathophysiology

- HbA = $\alpha_2\beta_2$ (HBA1, HBA2 code α ; **HBB codes β**).
- **HbS = glutamic acid → valine** at β -globin (A-for-T). **HbC = glutamic acid → lysine** (T-for-C).
- **Sickle-cell syndromes: SS, SC, S/ β^0 or S/ β^+ thalassemia, SE** — all ↑ pregnancy morbidity.
- Deoxygenation → sickling → membrane damage → irreversibly sickled cells; slowed microvascular transit (endothelial adhesion, dehydration, vasomotor dysregulation).

- **Clinical:** ischemia/infarction, **painful crises**; aplastic, megaloblastic, sequestration and hemolytic crises; avascular necrosis of femoral/humeral heads, renal medullary damage, **autosplenectomy (SS)** but splenomegaly in variants, cardiomegaly/ventricular hypertrophy, pulmonary infarction, **stroke**, leg ulcers, **infection/sepsis**. **Pulmonary hypertension in 20%** of adults with SS.
- **Treatment:** supportive care; **HbF induction with hydroxyurea** (↓ crises; teratogenic in animals, reassuring 17-yr follow-up); biologics, L-glutamine; stem cell transplant and **gene therapy** emerging as cures.

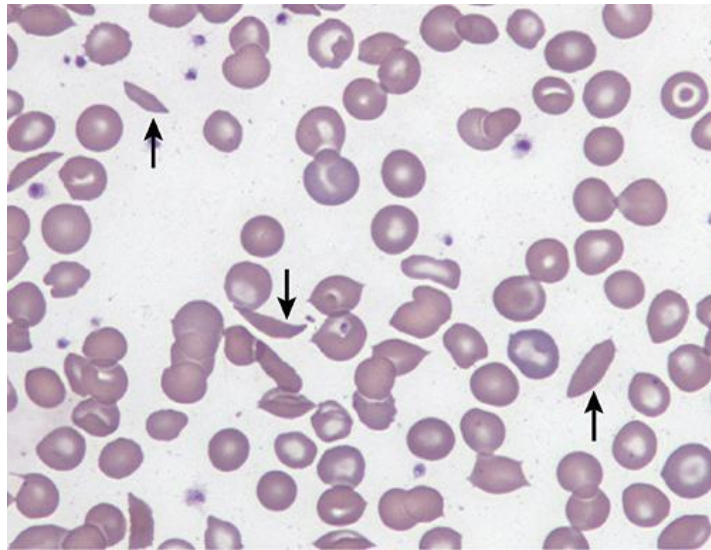


Figure 59-2. Sickle-cell anemia smear: elongated, crescent-shaped sickle cells (arrows).

Sickle-cell syndromes in pregnancy

- Serious burden, **especially SS**. Maternal mortality has improved; **perinatal morbidity/mortality remain formidable**.
- **Folic acid 4 mg/d**.

Table 59-3. Pregnancy Morbidity with Hemoglobin SS and SC Disease

Outcome	Hb SS (odds ratio)	Hb SC (odds ratio)
Preeclampsia	2–3.1	2.0
Stillbirth	6.5	3.2
Preterm delivery	2–2.7	1.5
Growth restriction	2.8–3.9	1.5
Maternal mortality	11–23	11

From Boafor, 2016; Oteng-Ntim, 2015.

Pain crisis

- ⚠️ **Diagnose crisis only after excluding other causes** — ectopic, **abruption**, pyelonephritis, appendicitis.
- Crises occur especially **late pregnancy, labor/delivery, early puerperium**.
- **IV fluids + prompt opioids**; nasal O₂; epidural may help.
- **Transfusion after pain begins does not dramatically relieve it**; prophylactic transfusions do prevent further crises.
- Narcotic habituation → ↑ neonatal abstinence syndrome.

Infection

- ↑ **Bacteriuria and pyelonephritis** → **screen and treat**. Endotoxin → rapid hemolysis + marrow suppression.
- Pneumonia (*S pneumoniae*) common. **Vaccinate: pneumococcal, H influenzae type B, meningococcal**.

Acute chest syndrome

- **Pleuritic chest pain, fever, cough, lung infiltrate, hypoxia**, ± bone/joint pain. **~6%** of pregnant women.

- **Four precipitants: infection (~half), marrow emboli, thromboembolism, atelectasis.**
- Mean stay **10.5 days**; **ventilation ~15%**; **mortality ~3%**.
- Transfusion: simple vs exchange (exchange no better in one study and used 4× blood); **ASH conditionally suggests exchange transfusion.**

Cardiac

- Ventricular hypertrophy, worsened by chronic hypertension; preeclampsia or infection may precipitate **heart failure**; consider pulmonary hypertension.
- 4352 pregnancies: **nondelivery admissions 63%**, hypertensive disorders **19% (1.8×)**, FGR **6% (2.9×)**, **cesarean 45% (1.7×)**.
- **Hb SC**: much lower morbidity; < half symptomatic before pregnancy; but bone pain and **pulmonary infarction/embolization** more common in pregnancy.

Prophylactic red cell transfusion

- **Most dramatic benefit on maternal morbidity.** Parkland protocol: keep **Hct >25%** and **HbS <60%**.
 - ↓ Maternal morbidity and admissions, but perinatal outcomes still poor (preterm birth, FGR).
- RCT (72 women): ↓ painful crises, no perinatal difference. Meta-analysis (12 studies): ↓ maternal mortality, pulmonary complications, perinatal mortality.
- Risks: **delayed hemolytic reaction up to 10%**, **alloimmunization ~1/4**, infection. No iron overload found on liver biopsy.
- **Consensus: individualize** → ~60% will need transfusion.

Fetal assessment

- Serial growth scans from the **mid-2nd trimester**; **weekly BPP/NST from 32–34 wk.**
- NSTs may be nonreactive during a crisis, recovering afterward. **Placental abnormalities in 69%.**

Labor and delivery

- Manage like **cardiac disease**; comfortable, not oversedated. **Conduction analgesia is ideal.**
- Crossmatched blood available; **transfuse if Hct <20%** and a difficult vaginal or cesarean delivery is expected.
- **Vaginal delivery; cesarean for obstetric indications.**
- **Contraception**: CDC rates combined hormonal methods, IUDs, implants and progestin-only methods as acceptable (benefits outweigh risks).

Sickle-cell trait

- **~8% of African Americans (1 in 12).** Hematuria, renal papillary necrosis, **hyposthenuria**; may speed ESRD progression.
- **Not** linked to ↑ abortion, perinatal mortality, LBW or preeclampsia (gestational hypertension ↑ in one study).
- **Definite: 2× asymptomatic bacteriuria and UTI.**
- Not a deterrent to pregnancy or hormonal contraception.

Hemoglobin C, CC, C-β-thalassemia

- **~2%** of African Americans heterozygous for HbC; **even homozygous CC is innocuous** — problematic only as **SC**.
- CC and C-β-thal: mild–moderate anemia, otherwise normal outcomes; give folate and iron.

Hemoglobin E

- **2nd most frequent Hb variant worldwide**; SE Asia (E-related genotypes in **36% of Cambodians, 25% of Laotians**).
- Homozygous EE: little/no anemia, **marked microcytosis**, target cells. **E trait**: only ↑ bacteriuria.
- **E-β-thalassemia**: common severe childhood anemia; **3× preterm birth and FGR** in pregnancy.

Newborn screening and prenatal diagnosis

- SS, SC, CC detectable by **cord blood electrophoresis**; **all newborns screened** (mandated in most states).
- **Risk arithmetic (African Americans):**
 - SS: $1/12 \times 1/12 \times 1/4 = 1 \text{ in } 576$
 - SC: $1/12 \times 1/40 \times 1/4 = 1 \text{ in } 2000$ (HbC gene 1 in 40)
 - S- β -thal: $1/12 \times 1/40 \times 1/4 = 1 \text{ in } 2000$ (β -thal minor 1 in 40)
- Prenatal diagnosis is **DNA-based on CVS or amniocytes** (targeted mutation analysis, PCR).

8. Thalassemia Syndromes

- Impaired synthesis/instability of a globin $\rightarrow$ ineffective erythropoiesis, hemolysis, anemia. Classified clinically as **transfusion dependent** or **nontransfusion dependent**.

α -Thalassemias

- **HBA1 and HBA2 on chromosome 16** $\rightarrow$ **4 α -genes** ($\alpha\alpha/\alpha\alpha$); γ genes also duplicated.
- **α^0 = both genes on one chromosome deleted ($—/\alpha\alpha$); α^+ = one gene deleted ($—\alpha/\alpha\alpha$ or $—\alpha/—\alpha$).**

Table 59-4. Genotypes and Phenotypes of α -Thalassemia Syndromes

Syndrome	Genotype	Phenotype
Normal	$\alpha\alpha/\alpha\alpha$	Normal
α^+ -Thalassemia heterozygote	$—\alpha/\alpha\alpha$ or $\alpha\alpha/—\alpha$	Normal; silent carrier
α^+ -Thalassemia homozygote ^a	$—\alpha/—\alpha$	α-Thalassemia minor —mild hypochromic, microcytic anemia
α^0 -Thalassemia heterozygote ^b	$—/\alpha\alpha$	α-Thalassemia minor —mild hypochromic, microcytic anemia
Compound heterozygous α^0/α^+	$—/—\alpha$	Hb H (β_4) disease with moderate to severe hemolytic anemia
Homozygous α -thalassemia	$—/—$	Hb Bart (γ_4) disease, hydrops fetalis

^aMore common in African Americans. ^bMore common in Asian Americans. The printed first column is headed "Genotype" (duplicating the second column); it lists the syndrome.

- **Why Hb Bart hydrops occurs in Asians but not in those of African descent:** Asians carry **cis deletions ($—/\alpha\alpha$)** $\rightarrow$ two carriers can produce $—/—$. People of African descent carry **trans deletions ($—\alpha/—\alpha$)** $\rightarrow$ cannot produce $—/—$. α -thal minor ~2% in African descent; HbH rare, Hb Bart disease unreported.
- **Pregnancy by number of deletions:**
 - **1 deletion (silent carrier):** $\pm$ mild microcytic anemia
 - **2 deletions (α -thal minor):** minimal–moderate hypochromic microcytic anemia; usually unrecognized, **no maternal consequence**; distinguishing $—/\alpha\alpha$ from $—\alpha/—\alpha$ **needs DNA analysis**. Newborn has Hb Bart at birth, not replaced by HbH.
 - **3 deletions (HbH disease, $—/—\alpha$):** newborn has Hb Bart + HbH + HbA, then develops hemolytic anemia; adults have moderate–severe anemia that **worsens in pregnancy**.
 - **4 deletions (Hb Bart disease, α -thal major):** Hb Bart (γ_4) has **very high O₂ affinity** $\rightarrow$ **stillbirth or hydrops**, death soon after birth.
- **Fetal screening for Hb Bart:** **cardiothoracic ratio at 12–13 wk**; myocardial performance (**Tei index**) changes precede hydrops.

β -Thalassemias

- **β -globin cluster ($\delta\gamma\beta$) on chromosome 11;** >150 point mutations. $\downarrow$ β -globin $\rightarrow$ excess α -globin precipitates $\rightarrow$ membrane damage.

- **β -thal minor (heterozygote):** HbA_2 ($\alpha_2\delta_2$) $>3.5\%$; HbF ($\alpha_2\gamma_2$) $>2\%$; no or mild-moderate hypochromic microcytic anemia.
- **β -thal major (Cooley anemia, homozygous):** intense hemolysis, transfusion dependence $\rightarrow$ **hemochromatosis** (often fatal). **Thalassemia intermedia** = moderate anemia.
- Treatments: stem cell transplant; thalidomide + hydroxyurea (**both contraindicated in pregnancy**); lentiviral gene therapy; **luspatercept**.
- **Pregnancy:**
 - **Iron and folate for all affected women.**
 - β -thal minor: mild anemia from **ineffective erythropoiesis, not hemolysis** (red cell survival normal, Fig 59-3); FGR association.
 - **β -thal major** (transfused + chelated): 63 pregnancies without serious complications; reasonably safe **if cardiac function normal. Transfuse to keep Hb ~ 10 g/dL** + fetal growth surveillance.

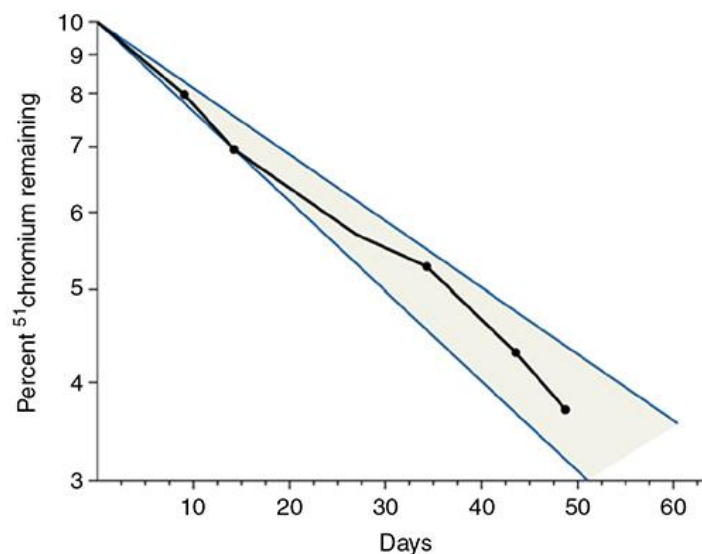


Figure 59-3. ^{51}Cr red cell survival in β -thalassemia minor (black line) lies within the normal pregnant range (shaded): the anemia is from ineffective erythropoiesis, not hemolysis.

Prenatal diagnosis of thalassemia

- **α -thal major:** DNA analysis; Hb Bart by capillary electrophoresis or HPLC. HBA1/HBA2 testing identifies **90% of deletions, 10% of point mutations**.
- **β -thal major:** many mutations $\rightarrow$ **familial mutation must be known first**; CVS; cell-free DNA methods described; **preimplantation genetic testing** possible.

9. Platelet Disorders

Thrombocytopenia

- **Platelets $<150,000/\mu\text{L}$ in $\sim 10\%$ of gravidas. 75% gestational thrombocytopenia;** HELLP is a common other cause. COVID-19 also causes it.

Table 59-5. Some Causes of Thrombocytopenia in Pregnancy

Cause
Gestational thrombocytopenia: 75 percent
Preeclampsia and HELLP syndromes: 20 percent
Obstetrical coagulopathies: DIC, MTP
Immune thrombocytopenic purpura
Systemic lupus erythematosus and APS
Infections: viral and bacterial
Drugs
Hemolytic anemias
Thrombotic microangiopathies
Malignancies
Pseudothrombocytopenia
Renal or liver diseases
Aplastic anemia
Genetic causes
COVID-19

APS = antiphospholipid syndrome; DIC = disseminated intravascular coagulopathy; HELLP = hemolysis, elevated liver enzyme levels, low platelet count; MTP = massive transfusion protocol. From ACOG, 2019c; Cooper, 2019; Tang, 2020.

Gestational thrombocytopenia

- 2.5th percentile **116,000–123,000/μL**; **~1% <100,000/μL**. Bleeding only with far lower counts.
- **Platelets <80,000/μL → look for another cause**; gestational thrombocytopenia is unlikely to be **<50,000/μL**.
- Nadir in the **3rd trimester**; hemodilution ± ↑ splenic mass.

Inherited thrombocytopenias

- **Bernard-Soulier syndrome, Glanzmann thrombasthenia**: missing platelet membrane glycoprotein → severe dysfunction. Mother can make antibodies to fetal **GP1b/IX** → **alloimmune fetal thrombocytopenia**.
 - 30 pregnancies: **primary PPH 33%** (half transfused); 6 neonatal alloimmune thrombocytopenias, 2 perinatal deaths. **Monitor through 6 wk postpartum**.
- **May-Hegglin anomaly**: autosomal dominant; thrombocytopenia, **giant platelets, leukocyte inclusions**; 75 pregnancies: 4 PPH, 34 neonatal thrombocytopenia, 2 fetal deaths.

Immune thrombocytopenic purpura (ITP)

- **IgG antibodies against platelet glycoproteins** → destruction in the reticuloendothelial system (**spleen**). Platelets **10,000–100,000/μL**.
- In adults usually **chronic**, rarely remits spontaneously.
- **Secondary**: SLE, lymphoma, leukemia, systemic disease; ~2% have positive lupus serology (± anticardiolipin); **~10% of HIV patients** are thrombocytopenic.

Table 59-6. Classification of Immune Thrombocytopenia (ITP)

Category	Definition
Etiology	
Primary ITP	Acquired immune-mediated disorder characterized by isolated thrombocytopenia in the absence of obvious initiating or underlying cause
Secondary ITP	Thrombocytopenia due to underlying cause/drug exposure
Duration	
Newly diagnosed	Persistent 3–12 months
Chronic	12 months or more

From Rodeghiero, 2009.

ITP in pregnancy

- **~1 in 10,000 births.** Pregnancy does **not** ↑ relapse or worsen active disease, though remitted women may recur (closer surveillance, hyperestrogenemia).
- **Treat if symptomatic bleeding and platelets <30,000/μL; target 50,000/μL.**
- **First line: corticosteroids and/or IVIG.**
 - **Prednisone 1 mg/kg/d** (suppresses splenic macrophages).
 - **IVIG 2 g/kg total over 2–5 days.**
 - Also used: azathioprine, rituximab; **TPO-receptor agonists (romiplostim, eltrombopag).**
- **Refractory → splenectomy** (open or laparoscopic; cesarean may be needed for exposure late in pregnancy); improvement in **1–3 days, peak ~8 days.**
- UKOSS (107 pregnancies): **PPH in half (20% severe)**; no neonate needed treatment; no ICH. Another series: 1/3 were overtreated.
- **Fetal/neonatal:**
 - ↑ Stillbirth, fetal loss, preterm birth. **IgG crosses the placenta.**
 - **Neonatal ICH <1%** (>800 neonates); **not related to route of delivery.**
 - **Maternal and fetal platelet counts correlate poorly**; antibody levels don't predict either.
 - **Fetal platelet sampling is NOT recommended.**
- **Alloimmune thrombocytopenia** (maternal antibodies to fetal platelet antigens) → Ch 18.

Thrombocytosis

- **>450,000/μL persistently.**
- **Reactive:** infection, iron deficiency, trauma, inflammation, malignancy — **seldom >1 million/μL.**
- **Essential thrombocythemia:** most cases **>1 million/μL**; clonal, often **JAK2**; arterial and venous thromboses.
 - First line: **aspirin ± hydroxyurea.**
 - Pregnancy: ~half had abortion, fetal death or preeclampsia (40 pregnancies); Mayo: 1/3 miscarried; **aspirin → abortion 1% vs 75% untreated.**
 - Pregnancy options: **aspirin, LMWH, interferon-α.**

10. Thrombotic Microangiopathies (TTP and HUS)

Etiopathogenesis

- **Thrombocytopenia + microangiopathic hemolysis + microvascular thrombosis.**

- **TTP:** antibodies to, or deficiency of, **ADAMTS13** (endothelial metalloprotease that cleaves vWF) → **ultralarge vWF multimers** → platelet aggregation → hyaline platelet–fibrin microthrombi in arterioles/capillaries → ischemia.
- **HUS:** endothelial damage from viral/bacterial infection; mainly **children**.
- **Secondary TMAs = 94%** — many pregnancy-related; also malignancy, drugs, transplant, autoimmune disease, COVID-19.

Manifestations

- **TTP pentad: thrombocytopenia, hemolytic anemia, fever, renal impairment, neurological abnormalities.**
- **HUS: more renal, fewer neurological** features.
- Severe thrombocytopenia but **spontaneous severe bleeding uncommon**.
- Hemolysis: moderate–marked anemia; **schizocytes/fragmented RBCs**, ↑ reticulocytes and nucleated RBCs, ↑ **LDH**, ↓ **haptoglobin**. Consumptive coagulopathy usually subtle.

Treatment

- **TTP: plasmapheresis with FFP replacement + glucocorticoids** (removes inhibitor, replaces ADAMTS13).
- **Caplacizumab** (anti-vWF) blocks ultralarge vWF–platelet interaction.
- RBC transfusion for life-threatening anemia. **Continue until platelets normal for 2 days**. Relapses common; renal sequelae.
- **Pregnancy-associated (complement-mediated) HUS: eculizumab** (anti-C5).

TMA in pregnancy

- **ADAMTS13 activity ↓ up to 50%** across pregnancy; lower still with preeclampsia and especially HELLP.
- **Parkland frequency 1 in 25,000** (11 in ~275,000); higher reported rates probably include severe preeclampsia.
- **Plasmapheresis is NOT for preeclampsia/HELLP. ⚠ Delivery reverses preeclampsia but does NOT improve TMA.** Be sure of the diagnosis before treating.
- ADAMTS13 is hard to interpret with HELLP.
- Mortality was up to half; much better with plasma exchange. TTP = **1%** of UK maternal deaths (2003–08); 2 deaths in 60 pregnancy-associated HUS.

Table 59-7. Some Differential Factors between HELLP Syndrome and Thrombotic Microangiopathies^a

Factor	HELLP Syndrome	Thrombotic Microangiopathies
Thrombocytopenia	Mild/mod.	Mod./severe
Microangiopathic hemolysis (schizocytosis)	Mild	Severe
ADAMTS13 deficiency	Mild/mod.	Severe
DIC	Mild	Mild
Transaminitis (AST, ALT)	Mod./severe	None/mild
Treatment	Delivery	Plasmapheresis

^aIncludes thrombotic thrombocytopenic purpura (TTP) and hemolytic uremic syndrome (HUS). ADAMTS13 = ADAM metalloprotease with thrombospondin type 1 motif, 13; AST = aspartate transaminase; ALT = alanine transaminase; DIC = disseminated intravascular coagulopathy; HELLP = hemolysis, elevated liver enzyme levels, low platelet count; Mod. = moderate.

Memory hook: "HELLP hits the Liver, TMA hits the Red cells." Big transaminases → HELLP (deliver); big hemolysis with little liver injury → TMA (plasmapheresis).

Recurrence and long-term prognosis

- Usually **recurrent** and often unrelated to pregnancy: 7 of 11 Parkland women recurred (nonpregnant or next 1st trimester). TTP recurred in 5 of 36 later pregnancies; atypical HUS in 5 of 32.
- **Relapse up to 40%** (nonpregnant). Long-term: recurrences, **dialysis/transplant**, severe chronic hypertension, transfusion-acquired infection. **Atypical HUS: 15% long-term renal failure**; 8 of 19 with primary aHUS reached ESRD.

11. Inherited Coagulation Defects

Hemophilias A and B

- **Hemophilia A = factor VIII deficiency; B (Christmas disease) = factor IX. X-linked recessive.**
- **Severity by factor level: mild 6–30%; moderate 2–5%; severe <1%.**
- Women: usually **carriers** with ↓ levels; true disease almost needs homozygosity (or new mutation). **Acquired hemophilia A** (antibodies) in pregnancy → severe bleeding.
- **Pregnancy:**
 - Bleeding risk ∝ factor level. Carrier levels average **~50%** (lyonization); **<10–20% → hemorrhage risk.**
 - Factor VIII and IX **rise in pregnancy** (protective).
 - **Desmopressin** releases factor VIII; avoid lacerations, minimize episiotomy, maximize uterine contraction; operative and cesarean delivery carry bleeding risk.
 - Carriers: **PPH 20%**. Hemophilia B: **give factor IX if <10%**.
 - Affected male neonate: bleeding risk — especially **circumcision**.
- **Inheritance (carrier mother): half of sons affected, half of daughters carriers.** (Two abnormal alleles: all sons affected, all daughters carriers.) Prenatal diagnosis by CVS; PGT.

The chapter text says a one-allele carrier "will have all sons affected ... and half of her daughters will be carriers"; for X-linked recessive inheritance with an unaffected father this should be half of her sons affected.

Factor VIII or IX inhibitors

- Rare acquired antibodies; in nonhemophiliacs **extraordinary**, but reported in the **puerperium**.
- **Severe, protracted, repetitive genital-tract bleeding starting ~1 week after an uncomplicated delivery; aPTT markedly prolonged.**
- Treatment: blood components, immunosuppression, surgery (curettage, hysterectomy); **recombinant factor VIIa (NovoSeven) stops bleeding in up to 75%.**

Von Willebrand disease (vWD)

- **Most common inherited bleeding disorder: up to 1–2%. ≥20 variants; mostly autosomal dominant; type I = 75%; type III most severe, recessive.**
- **vWF:** large multimeric glycoprotein from **endothelium and megakaryocytes**; mediates **platelet adhesion to subendothelial collagen** and **stabilizes factor VIII** (factor VIII is made in the liver).
- **Symptoms:** easy bruising, epistaxis, mucosal bleeding, **heavy menses**, bleeding with surgery/trauma.
- **Labs:** prolonged bleeding time and PTT; ↓ vWF antigen; ↓ factor VIII activity; abnormal platelet responses.
- Some advise **cesarean with severe maternal disease** to protect a possibly affected fetus.
- **Pregnancy:**
 - **Factor VIII and vWF rise** → often normalize (bleeding time may stay prolonged).
 - Treat if factor VIII very low or bleeding: **desmopressin** (transient ↑ VIII and vWF); significant bleeding → **cryoprecipitate 15–20 units every 12 h** or **vWF-containing factor VIII concentrates (Alphanate, Humate-P).**
 - **Neuraxial analgesia safe once coagulation corrected** or with prophylactic hemostatic agents.

- **PPH common:** 33% (systematic review, >800 deliveries) vs ~5% (database, >2200); may be **primary or delayed**.

Other factor deficiencies

Deficiency	Inheritance	Key pregnancy points
Factor VII	Autosomal recessive (rare)	Rises only mildly in pregnancy; rFVIIa prophylaxis did not change PPH (94 births)
Factor X (Stuart-Prower)	Autosomal recessive (rare)	Rises ~50% in pregnancy yet ↑ preterm birth, perinatal death, PPH; treat with plasma-derived factor X, FFP or PCC
Factor XI	Autosomal recessive; Ashkenazi Jews	70% uneventful; factor XI concentrate for cesarean; avoid epidural unless factor XI given ; levels correlate poorly with bleeding; PPH 18%
Factor XII	Autosomal recessive	Rare in pregnancy; ↑ thromboembolism (not bleeding)
Factor XIII	Autosomal recessive	Maternal ICH, recurrent miscarriage, abruption , umbilical cord bleeding; FFP; weekly factor XIII concentrate prophylaxis → successful pregnancies
Fibrinogen	Dysfibrinogenemia (autosomal); hypo-/afibrinogenemia (recessive; Parkland view: hypofibrinogenemia = heterozygous dominant)	Clottable protein 80–110 mg/dL nonpregnant; fibrinogen or plasma infusions through pregnancy

Thrombophilias

- Inherited deficiencies of coagulation inhibitors → recurrent thromboembolism → see **Ch 55**.

Quick-Recall: Differential Table

Condition	Key lab clue	Key management
Iron deficiency	MCV <80, ferritin <10–15	Elemental iron 200 mg/d ; IV iron sucrose if intolerant
Anemia of chronic disease	Microcytic, ferritin high , hepcidin ↑	Treat cause; EPO at Hct ~20% in renal disease
Folate deficiency	Hypersegmented neutrophils, macrocytes	Folic acid 5–15 mg/d + iron; retics ↑ by 4–7 d
B12 deficiency	Megaloblastic; after gastrectomy	1000 µg IM monthly
Autoimmune hemolysis	Coombs + (warm 80–90%)	Rituximab 100 mg weekly × 4 + prednisone
PNH	Complement-mediated hemolysis, VTE 40%	Eculizumab , anticoagulation
Hereditary spherocytosis	Spherocytes, ↑ osmotic fragility	Folic acid 4 mg/d; watch for parvovirus
G6PD deficiency	Episodic hemolysis with infection/drugs	Avoid triggers (e.g., nitrofurantoin)
Sickle-cell disease (SS)	Sickle cells; maternal mortality OR 11–23	Folate 4 mg, exclude other causes of pain, individualized transfusion (Hct >25%, HbS <60%)
α-thal minor vs β-thal minor	Normal HbA ₂ (DNA needed) vs HbA₂ >3.5% , HbF >2%	Iron + folate; partner testing
Hb Bart disease	—/—; cardiothoracic ratio at 12–13 wk	Hydrops; prenatal DNA diagnosis
Gestational thrombocytopenia	Mild, 3rd trimester, rarely <50,000	None; investigate if <80,000
ITP	Isolated thrombocytopenia, chronic	Treat if bleeding and <30,000 ; prednisone 1 mg/kg or IVIG 2 g/kg
TTP/HUS	Severe hemolysis, low ADAMTS13, mild liver	Plasmapheresis (TTP) ; eculizumab (aHUS) ; delivery doesn't help
HELLP	High transaminases, mild hemolysis	Delivery
vWD	↓ vWF, ↓ VIII, prolonged bleeding time	Desmopressin; cryo or Humate-P; watch for delayed PPH
Acquired factor VIII inhibitor	Bleeding ~1 wk postpartum, aPTT very long	rFVIIa (75% response), immunosuppression

Key Numbers to Memorize

- Anemia: Hb **<11 (1st/3rd) / <10.5 (2nd) g/dL**; Parkland Hct **<30%**; prevalence **3–38%**
- Iron need **~1000 mg**; prophylaxis **30–60 mg** iron + **400 µg** folate; treatment **200 mg/d**; ferritin **<10–15** diagnostic
- Stable woman with Hb **~7 g/dL** → oral iron **≥3 months**, not transfusion
- Folate need **400 µg/d**; treatment **5–15 mg/d**; response in **4–7 d**; B12 **1000 µg IM monthly** after total gastrectomy
- Warm AIHA **80–90%**; PNH VTE **40%**, stillbirth on eculizumab **4%**
- G6PD: **2%** homozygous, **10–15%** heterozygous African-American women; pyruvate kinase deficiency **1 in 10,000**
- Sickle trait **1 in 12 (8%)**; SS **1 in 576**; SC and S-β-thal **1 in 2000**; HbC gene **1 in 40**
- SS odds ratios: stillbirth **6.5**, maternal mortality **11–23**; acute chest **6%** (ventilation **15%**, mortality **3%**); pulmonary hypertension **20%**
- Transfusion targets: Hct **>25%**, HbS **<60%**; transfuse before difficult delivery if Hct **<20%**; alloimmunization **~25%**; **~60%** need transfusion

- β -thal minor: HbA₂ **>3.5%**, HbF **>2%**; β -thal major: keep Hb **~10 g/dL**
- Thrombocytopenia **<150,000** in **10%** (75% gestational, 20% preeclampsia/HELLP); investigate **<80,000**
- ITP: **1 in 10,000**; treat **<30,000** + bleeding; target **50,000**; prednisone **1 mg/kg**, IVIG **2 g/kg**; neonatal ICH **<1%**
- Thrombocytosis **>450,000**; essential **>1 million**; aspirin → abortion **1% vs 75%**
- TMA: **1 in 25,000**; ADAMTS13 ↓ **50%** in pregnancy; secondary TMA **94%**; relapse **40%**
- Hemophilia severity: mild **6–30%**, moderate **2–5%**, severe **<1%**; carriers PPH **20%**; factor IX if **<10%**
- vWD prevalence **1–2%**, type I **75%**; cryo **15–20 units q12h**; PPH **5–33%**; rFVIIa for inhibitors **75%**